

Web Development Research Document

Projects: Online Store and Loaning Website

Prepared for: Mr. Leonard Shivute

Prepared by: Stephanus Sylvanus

Date: 31 July 2025

1. Introduction

This document provides a comprehensive overview of the development approach for two web applications: an Online Store and a Loaning Website. The goal is to deliver responsive, secure and user-friendly web platforms using modern development tools and technologies.

2. Project Requirements

2.1 Online Store

- Product catalog with descriptions and pricing
- User registration and login
- Integration with WhatsApp for order placement
- Admin panel to manage products

2.2 Loaning Website

- User-friendly loan application form
- Admin dashboard for application review
- Authentication and document upload support
- Notification via WhatsApp or email

3. Tools & Technologies Researched

Layer	Tools Considered	Chosen Tool	Reason for Selection
Frontend	HTML, CSS, JS, React, Vue, Bootstrap	React.js	Provides a scalable and efficient way to build interactive UIs.
Styling	Bootstrap, Tailwind CSS, Vanilla CSS	CSS	Direct control over styling and layout customization.

Backend	Firestore, Node.js, Express.js, PHP	Firestore	Serverless backend solution with real-time support and scalability.
Database	Firestore, Supabase, MongoDB	Supabase	SQL-based, open-source backend with real-time capabilities.
Authentication	Firestore Auth, Auth0, Supabase Auth	Supabase	Integrated auth system that works seamlessly with Supabase DB.
Payments	Stripe, PayPal, PayFast, WhatsApp	WhatsApp Redirect	Aligns with business preference to handle orders via WhatsApp.
Design	Adobe XD, Figma	Figma	Free, collaborative, and cloud-based design tool.
Development Tools	VS Code, Sublime Text, WebStorm	VS Code	Popular, lightweight IDE with broad extension support.

4. Final Technology Stack

Component	Technology Used
Frontend	HTML, CSS, JavaScript, React.js
Backend	Firestore
Database	Supabase (PostgreSQL)
Authentication	Supabase Auth
Payment Gateway	WhatsApp redirect link
Design Tools	Figma
Code Editor	Visual Studio Code

5. Development Workflow

1. Design UI mockups using Figma.
2. Set up frontend with React.js and basic routing.
3. Create reusable components and style using CSS.
4. Integrate Firebase for backend services and hosting.
5. Connect Supabase for storing and retrieving user and application data.
6. Implement Supabase authentication for secure access.
7. Add WhatsApp redirect functionality for orders.
8. Test all features across browsers and devices.
9. Deploy to Firebase Hosting.

6. References

<https://reactjs.org>

<https://supabase.com/docs>

<https://firebase.google.com/docs>

<https://figma.com>

<https://code.visualstudio.com>